

Project Description

BAP Service Time Prediction — Puerto de San Antonio, Chile

1. Objective

Predict the service time of vessels at Puerto de San Antonio to support a Berth Allocation Problem (BAP) optimizer. The port has 5 terminals (DP World, STI, PANUL, QC, EPSA) that independently plan berth schedules but share a common port entry channel. Accurate service time predictions enable better berth scheduling and port entry coordination.

2. Problem Context

The BAP assigns incoming vessels to available berths while minimizing waiting times and maximizing utilization. A key input is the expected service time: how long each vessel will occupy a berth. This project builds a training dataset to develop a predictive model for that service time, using only information available before the vessel berths.

Port Operations Sequence

- Arrival at anchorage:** F. arribo — vessel arrives and anchors outside the port
- Waiting at anchorage:** Vessel waits for berth availability (can be days for bulk carriers)
- Pilot boarding:** Fecha práctico atraque — pilot boards, vessel enters port channel
- Mooring:** 1era espía atraque to Última espía atraque — ~42 min average
- Cargo operations:** Loading/unloading at berth
- Unmooring:** 1era espía desatraque to Última espía desatraque
- Departure:** Zarpe — vessel exits the port

Key insight: F. arribo records arrival at the anchorage (outside the port), not entry into the port. Dry bulk vessels wait a median of 207 hours (~8.6 days) at anchor before the pilot boards. The pilot-to-mooring gap is a consistent ~42 minutes across all vessel types, confirming pilot boarding marks actual port entry.

3. Target Variables

Target	Formula	Use Case
estadia_sitio_hours (Berth stay)	Última espía desatraque – 1era espía atraque	Berth allocation: how long the berth is occupied
tiempo_en_puerto_hours (Port time)	Zarpe – Fecha práctico atraque	Channel scheduling: how long the vessel is inside the port

4. Data Source

File: BBDD limpia(1).xlsx, sheet "Resume Naves Comerciales (4)"

Records: 5,605 vessel calls (5,597 after cleaning)

Period: December 2019 to August 2025

Completeness: All timestamp and numeric columns are 100% complete. Only shipping line and service route have missing values (~30%), concentrated in non-container vessel types.

5. Feature Engineering

Features use only pre-berthing information (no data leakage):

Direct features: vessel type, gross tonnage, terminal, agency, shipping line, arrival drafts, origin/destination ports and regions

Temporal features: month, day of week, hour, year, quarter, weekend flag

Historical features: expanding-window statistics of the same vessel's prior berth stays (mean, median, std, last visit, count), with shift-by-1 to prevent leakage

Group-level features: expanding-window averages at (vessel type, terminal), vessel type, and terminal levels

Berth assignment (Sitio) is excluded as it is the BAP decision variable. Departure drafts are excluded as they are not known pre-berthing.

6. Vessel Type Grouping

The 16 original vessel types are aggregated into 7 operationally meaningful groups based on cargo handling methods and service time patterns:

Group	Original Types	Count	Avg Stay (h)
Container	Contenedor	3,066	36
Dry Bulk	Carga Seca Granel, Mineral/Granel/Petrolero	951	74
Vehicle Carrier	Autero, Autotrasbordo	749	36
Liquid Bulk	Transp. Quimico/Liquido/Asfalto, Petrolero	587	22
General Cargo	Tradicional, C. Proyecto, Chipero, Refrig., Otros	171	36
Passenger	Pasajeros	80	19
Other	Nave Armada	1	65

7. Data Cleaning

- 7 records with berth stay < 2 hours removed (aborted calls or data errors)
- 1 record with berth stay of 780 hours removed (anomalous)
- 12 records flagged where dispatch timestamp precedes reception (quality issue, targets remain valid)

8. Suggested Train/Test Split

A temporal split prevents data leakage from the historical features:

Split	Period	Records
Train	Before 2024-07-01	4,413
Validation	2024-07-01 to 2025-02-28	688
Test	After 2025-03-01	496

9. External Data Enrichment

External datasets were collected to supplement the main vessel call database. All external data is stored in the `external_data/` folder with full source documentation in `sources.md`.

Weather Data (Open-Meteo API, CC BY 4.0)

- weather_daily.csv: 2,056 days of temperature, precipitation, wind speed/gusts/direction (complete)
- weather_hourly.csv: 49,344 hours of wind and precipitation data (complete)
- marine_weather_daily.csv: 2,056 days of wave height, period, swell (31% null before Oct 2021)

Weather is the most impactful enrichment: wind and wave conditions directly affect mooring operations, port entry, and can extend service times.

Vessel Characteristics

vessel_dimensions.csv: LOA (length overall) for 338 of 1,550 vessels (22%), derived by cross-matching IMO codes through imo_vessel_codes.csv (64K vessels, GitHub, Public Domain) and vessel_information_ais.csv (10K vessels with LOA). Fuller coverage requires Kaggle account or paid databases (Datalastic, MarineTraffic).

Reference Datasets

- port_noumea_stopovers.csv: ~50K port calls from Noumea (2002-2017), 35 vessels overlap with our fleet (Zenodo)
- ship_movement_2015-2022.zip: Berth mooring dynamics, 14 ships with length/beam/DWT and weather (Zenodo, CC BY)

10. Known Limitations

- No cargo volume data in the main database. Container counts exist only for ~90 vessels in stats plan 2025. Draft difference could proxy for volume but is only available post-berthing.
- No crane/equipment data. Cannot normalize service time by handling resources.
- Vessel dimensions (LOA, beam, DWT) available for only 22% of vessels from free sources.
- 30% missing shipping line/service data for non-container types. Structurally missing (bulk/tanker vessels don't use liner services).
- Weather data not yet joined to training dataset. Available in external_data/ for integration.
- Marine wave data has gaps before October 2021 (31% null).

11. Project Structure

File	Description
data/BDD limpia(1).xlsx	Source data (unchanged)
data/training_dataset.csv	Generated training dataset (5,597 rows x 44 cols)
data/column_description.pdf	Column definitions (EN/ES)
data/project_description.pdf	This document
external_data/	Weather, vessel dimensions, reference datasets + sources.md
src/build_training_dataset.py	Dataset builder script
src/download_external_data.py	External data downloader
src/port_regions.py	Port-to-region mapping dictionary
src/generate_pdfs.py	PDF documentation generator